



US012407972B2

(12) **United States Patent**
Marti et al.

(10) **Patent No.:** **US 12,407,972 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **AUDITORY AUGMENTED REALITY USING
SELECTIVE NOISE CANCELLATION**

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(*) Notice: Subject to any disclaimer, the term of this
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(21) Appl. No.: **17/672,461**

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(22) Filed: **Feb. 15, 2022**

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(65) **Prior Publication Data**

US 2022/0174395 A1 Jun. 2, 2022

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(57) **ABSTRACT**

Techniques for auditory augmented reality using selective noise cancellation include receiving an input signal capturing an ambient auditory environment; separating the input signal into a set of audio signals that includes first and second component signals; and in response to generating the set of audio signals: generating a context-sensitive user interface that displays a plurality of first controls for modifying the first component signal and a plurality of second controls for modifying the second component signal, the plurality of first controls being independent from the plurality of second controls; receiving, from a user, a selection to remove the first component signal; and in response to receiving the selection: removing the first component signal from the set of audio signals to generate a modified set of audio signals that includes the second component signal; and driving an audio output device to generate sound based on the modified set of audio signals.

Related U.S. Application Data

(63) Continuation of application No. 16/907,063, filed on Jun. 19, 2020, now Pat. No. 11,284,183.

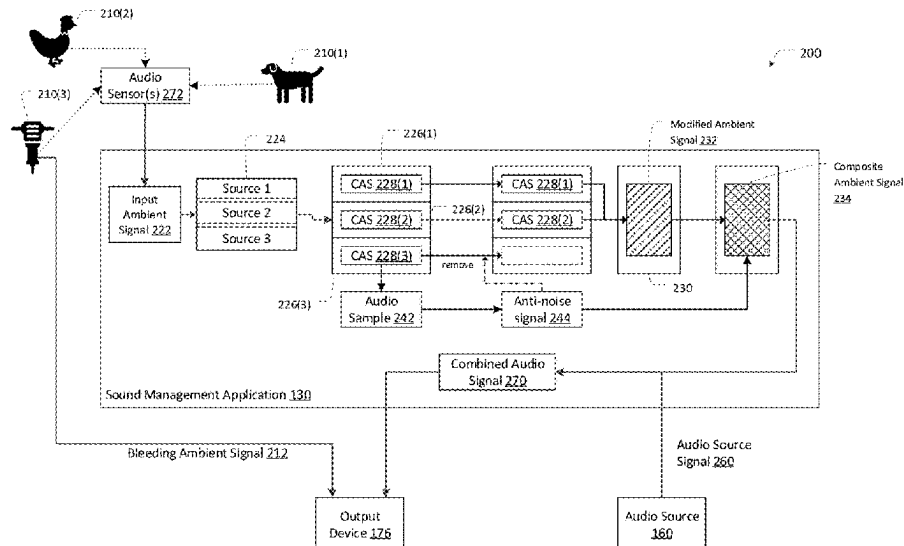
(51) **Int. Cl.**
G10K 11/178 (2006.01)
H04R 1/10 (2006.01)

(52) **U.S. Cl.**
CPC **H04R 1/1083** (2013.01); **G10K 11/178**
(2013.01); **H04R 2460/01** (2013.01)

(58) **Field of Classification Search**
CPC H04R 1/1083; H04R 2460/01; G10K
11/178; G10K 2210/1081;

(Continued)

20 Claims, 5 Drawing Sheets



(58) **Field of Classification Search**

CPC G10K 2210/3025; G10K 11/17823; G10K
 11/17873; G10K 11/17885; G10K
 2210/30231
 USPC 381/1
 See application file for complete search history.

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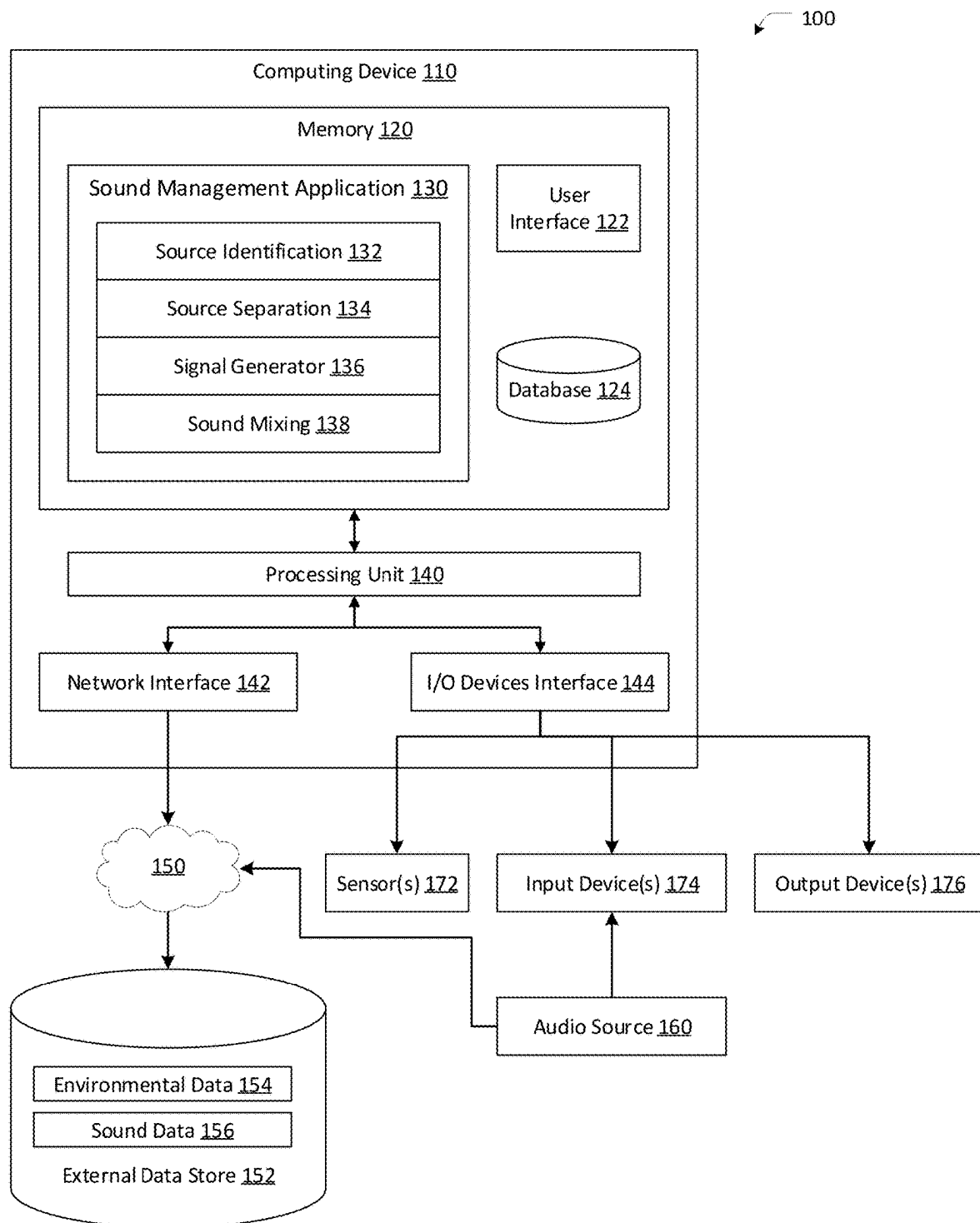


FIGURE 1

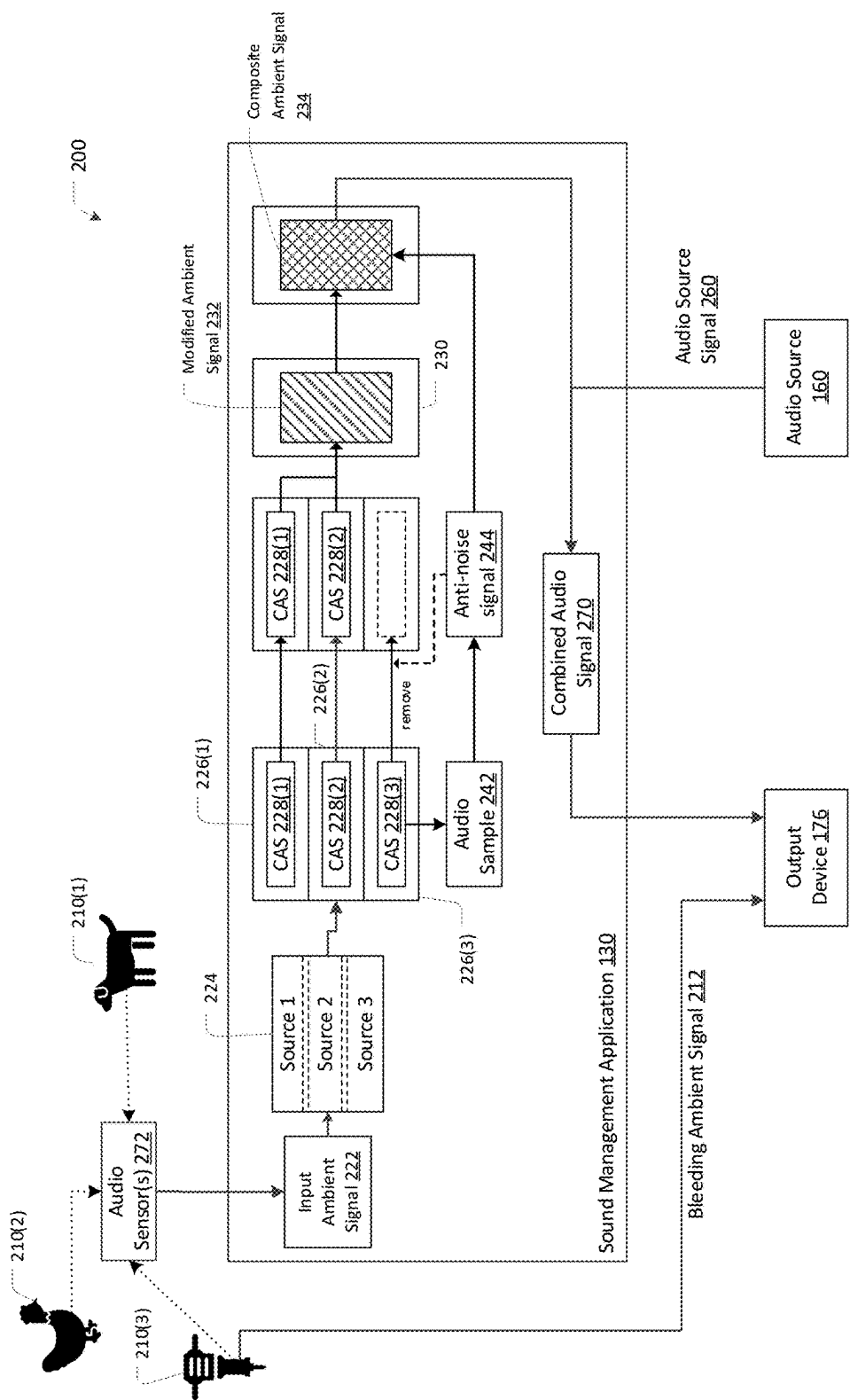


FIGURE 2

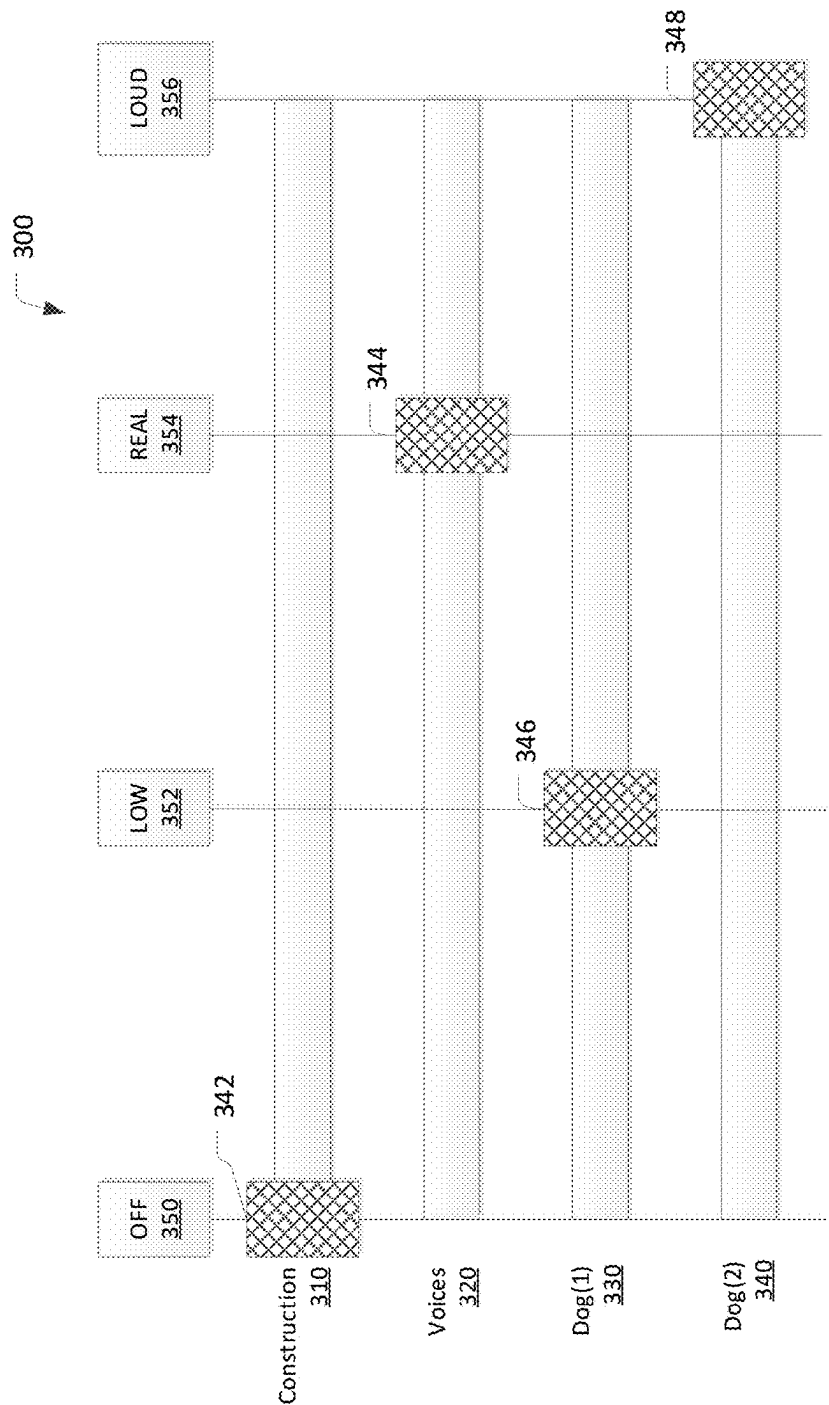


FIGURE 3

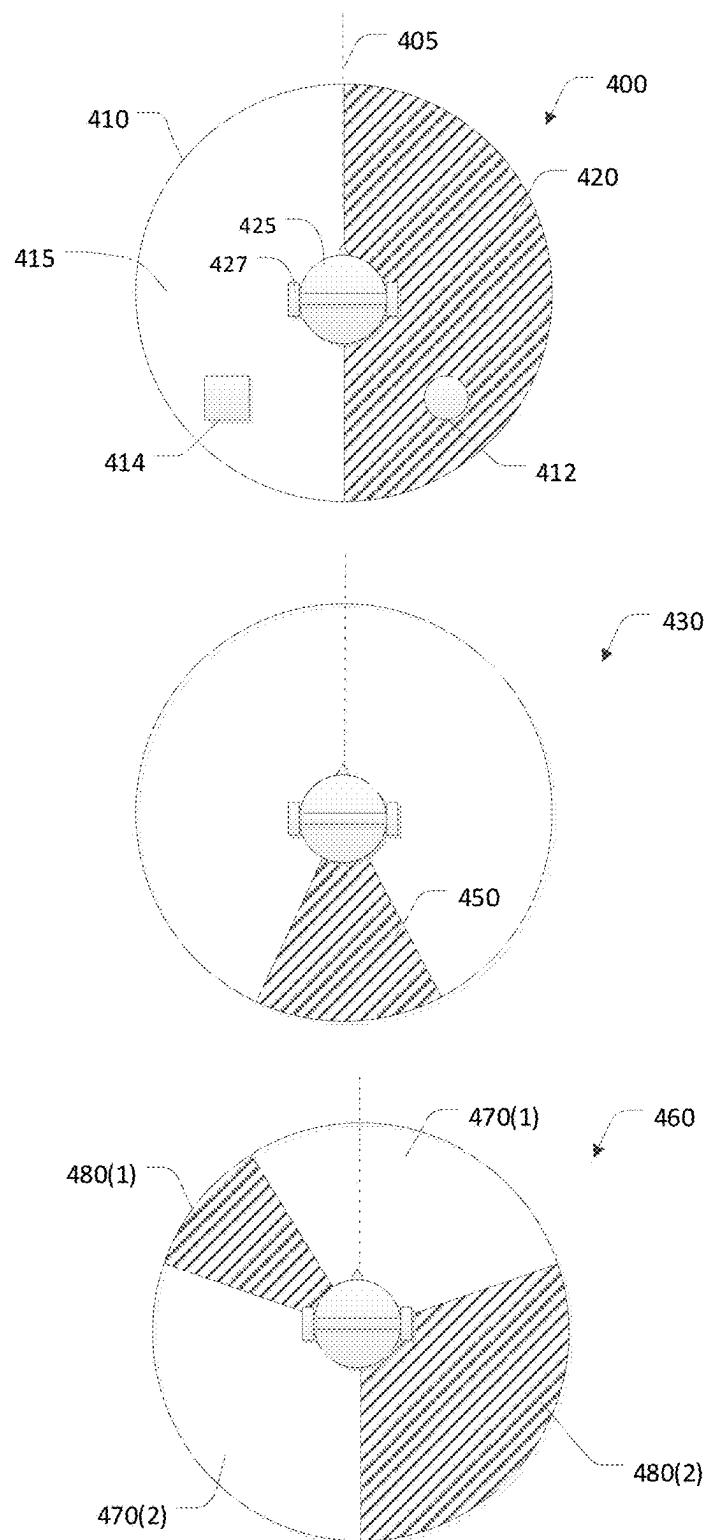
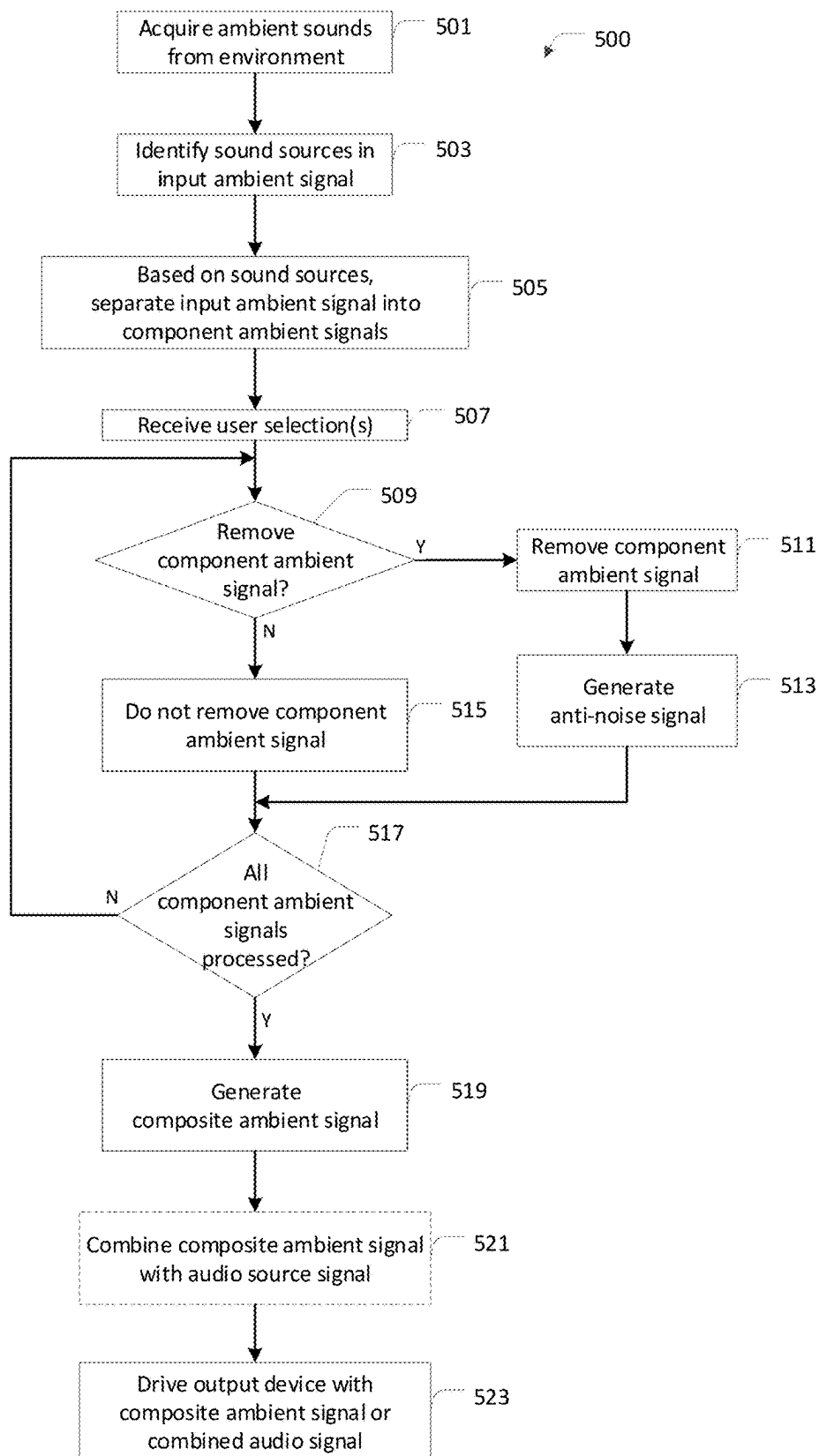


FIGURE 4

**FIGURE 5**

1

AUDITORY AUGMENTED REALITY USING SELECTIVE NOISE CANCELLATION

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of United States application titled "AUDITORY AUGMENTED REALITY USING SELECTIVE NOISE CANCELLATION," filed on Jun. 19, 2020, and having Ser. No. 16/907,063. The subject matter of this related application is hereby incorporated herein by reference.

BACKGROUND

Field of the Various Embodiments

Embodiments disclosed herein relate to sound modification and, in particular, auditory augmented reality using selective noise cancellation.

Description of the Related Art

Various audio products employ techniques to remove or attenuate unwanted sounds included in an ambient environment such that a user can listen to a desired audio source. In many cases, headphones generally provide a degree of passive noise attenuation by fully or partially obstructing the ear canal of the wearer. Additionally, some headphones provide active noise attenuation by generating sound waves that cancel sounds within the environment. Some conventional devices also attempt to provide the sounds of the ambient environment by recording the ambient environment and including the recording in the sound reproduced by an audio output device. In such instances, the output device is "acoustically transparent," meaning that the audio product does not alter the auditory field or environment experienced by a user.

One drawback of these conventional approaches is that, by canceling all sound in the ambient environment, the user is isolated from sounds in the surrounding environment. When the user is isolated from sounds in the surrounding environment, the user can miss sounds that may be of interest to the user, such as speech from other people (e.g., announcements at an airport, someone calling for the user). Alternatively, when providing sounds of interest, conventional approaches reproduce then entire ambient environment, exposing the user to undesirable noise and detracting from the overall enjoyment of the listening experience.

As the foregoing illustrates, what is needed are more effective techniques for noise attenuation.

SUMMARY

Various embodiments include a computer-implemented method comprising receiving an input signal representing an ambient auditory environment of a user, generating, from the input signal, a set of ambient audio signals that includes a first component signal and a second component signal, generating, based on the first component signal, a first inverse signal that is a polar inverse of the first component signal, removing the first component signal from the set of ambient audio signals, generating a first composite signal that includes at least the first inverse signal and the second component signal, and driving an audio output device to produce soundwaves based on the first composite signal.

2

Further embodiments provide, among other things, a system and a non-transitory computer readable storage medium configured to implement the method set forth above.

At least one technological advantage of the disclosed approach relative to the prior art is that by executing active noise cancellation on selected sounds, the disclosed approach enables a system to prevent a user from hearing particular sounds by removing a selected sound from a reproduced ambient acoustic environment, as well as by cancelling bleeding sounds that are not removed using other noise cancellation techniques. These technical advantages provide one or more technological advancements over prior art approaches.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the various embodiments can be understood in detail, a more particular description of the inventive concepts, briefly summarized above, may be had by reference to various embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of the inventive concepts and are therefore not to be considered limiting of scope in any way, and that there are other equally effective embodiments.

FIG. 1 illustrates a transparent sound modification system according to one or more embodiments;

FIG. 2 illustrates a technique of the transparent sound modification system of FIG. 1 applying active noise cancellation to a portion of an ambient audio signal, according to one or more embodiments;

FIG. 3 illustrates an example user interface provided by the transparent sound modification system of FIG. 1 to control portions of an ambient audio signal, according to one or more embodiments;

FIG. 4 illustrates selection of directions within an environment for the transparent sound modification system of FIG. 1 to modify, according to one or more embodiments; and

FIG. 5 is a flow diagram of method steps to apply active noise cancellation to a portion of an ambient audio signal, according to one or more embodiments.

DETAILED DESCRIPTION

In the following description, numerous specific details are set forth to provide a more thorough understanding of the various embodiments. However, it will be apparent to one of skilled in the art that the inventive concepts may be practiced without one or more of these specific details.

Overview

Embodiments disclosed herein include a transparent sound modification system that includes one or more sensors arranged to detect sounds within an environment, and one or more output devices that provide a modified audio signal to a user. A processing unit included in the transparent sound modification system operates to detect sounds within the environment and generate a modified audio signal that attenuates or removes at least one of the detected sounds and to generate a modified ambient signal. The processing unit also provides active noise cancellation for the detected sound that is attenuated or removed by generating an inverse sound signal based on the detected sound. The processing

unit combines the inverse sound signal with the modified ambient signal to generate a composite ambient signal. The processing unit combines the composite ambient signal with an audio source signal to generate a combined audio signal. In instances where a removed sound nonetheless leaks through the audio output device and would otherwise be heard by the user as a bleeding sound, the inverse sound signal included in the modified ambient signal is reproduced by the output device and actively cancels the bleeding sound. The user therefore hears a combined signal that includes only the audio source signal and the portions of the detected sounds that were not attenuated or removed.

The transparent sound modification system may be implemented in various forms of audio-based systems, such as personal headphones, home stereo systems, car stereo systems, etc. The transparent sound modification system may selectively provide active noise attenuation for selected sound sources. The transparent sound modification system may perform its processing functions using a dedicated processing device and/or a separate computing device, such as a user's mobile computing device or a cloud computing system. The transparent sound modification system may detect sounds from the environment using any number of sensors, which may be attached to or integrated with other system components, or disposed separately. The detected sounds, location information, and user inputs, such as selected sound sources or selected sound source directions, may be used to separate, isolate, and remove specific sounds in the environment while transmitting other sounds within the environment in order to provide user-configurable acoustic transparency.

System

FIG. 1 illustrates a transparent sound modification system according to one or more embodiments. As shown, transparent sound modification system 100 includes computing device 110, network 150, external data store 152, audio source 160, sensor(s) 172, input device(s) 174, and output device(s) 176. Computing device 110 includes memory 120, processing unit 140, network interface 142, and input/output devices interface 144. Memory 120 includes user interface 122, database 124, and sound management application 130.

Computing device 110 includes processing unit 140 and memory 120. In various embodiments, computing device 110 may be a device that includes one or more processing units 140, such as a system-on-a-chip (SoC). In some embodiments, computing device 110 may be a wearable device, such as headphones, hearing aids, portable speakers, and/or other devices that include processing unit 140. In other embodiments, computing device 110 may be a mobile computing device, such as a tablet computer, mobile phone, media player, and so forth. In some embodiments, computing device 110 may be a head unit included in a vehicle system or at-home entertainment system. Generally, computing device 110 can be configured to coordinate the overall operation of transparent sound modification system 100. The embodiments disclosed herein contemplate any technically-feasible system 100 configured to implement the functionality of transparent sound modification system 100 via computing device 110.

In various embodiments, one or more of computing device 110, sensor(s) 172, input device(s) 174, and/or output device(s) 176 may be included in one or more devices, such as mobile devices (e.g., cellphones, tablets, laptops, etc.), wearable devices (e.g., watches, rings, bracelets, headphones, etc.), consumer products (e.g., gaming, gambling, etc.), smart home devices (e.g., smart lighting systems, security systems, digital assistants, etc.), communications

systems (e.g., conference call systems, video conferencing systems, etc.), and so forth. Computing device 110 may be located in various environments including, without limitation, road vehicle environments (e.g., consumer car, commercial truck, etc.), aerospace and/or aeronautical environments (e.g., airplanes, helicopters, spaceships, etc.), nautical and submarine environments, and so forth.

For example, a wearable device could include at least one microphone as sensor 172, at least one speaker as output device 176, and a microprocessor-based digital signal processor (DSP) as processing unit 140 that produces audio signals that drive the at least one speaker to emit sound-waves. In some embodiments, transparent sound modification system 100 may be included in headphones with individual earpieces, or ear bud sets, where each earpiece or ear bud contains a speaker, one or more microphones, and/or one or more transducers. In such instances, the one or more microphones may include an ambient microphone (e.g., sensor 172(1)) that detects ambient sounds originating from sound sources within the ambient auditory environment, and an internal microphone (e.g., sensor 172(2)) used in a closed-loop feedback control system for cancellation of user-selected sounds from the ambient auditory environment.

In various embodiments, computing device 110 may operate in a processing mode that drives output device(s) 176 such that output device(s) 176 are, at least in part, acoustically transparent, meaning transparent sound modification system 100 does not alter the environment experienced by the user relative to the current ambient auditory environment. Providing acoustic transparency causes transparent sound modification system 100 to refrain from altering one or more source sounds included in the ambient auditory environment. For example, computing device 110 could drive speakers 176 included in a pair of headphones to reproduce the ambient auditory environment as an audio signal.

In various embodiments, computing device 110 may attenuate a portion of sounds, such as particular portions associated with one or more source sounds in the ambient auditory environment. In such instances, transparent sound modification system 100 analyze the ambient auditory environment to identify one or more sound sources and may provide controls associated with each of the sound sources and/or sound types. The user may then customize her auditory environment by selecting specific sound sources and/or specific sound types to reproduce, amplify, attenuate, and/or remove from the ambient auditory environment. In such instances, transparent sound modification system 100 may separate signals based on the set of sounds. In various embodiments, transparent sound modification system 100 may, through a combination of passive and/or active attenuation or cancellation, block the user from hearing sounds associated with the selected sound sources and/or sound types. At the same time, transparent sound modification system 100 may reproduce a composite ambient signal that includes only sounds associated with the sound sources and/or sound types that the user selected for full reproduction and/or amplification.

In alternative embodiments, various components of transparent sound modification system 100 may be contained within, or implemented by, different kinds of wearable devices and/or non-wearable devices. For example, one or more of computing device 110, sensor(s) 172, input device(s) 174, and/or output device(s) 176 may be disposed within a hat, scarf, shirt collar, jacket, hood, etc. Similarly, processing unit 140 may provide the user interface via output

device(s) **176** included in a separate mobile or wearable device, such as a smartphone, tablet, wrist watch, arm band, etc. The separate mobile or wearable device may include an associated microprocessor and/or a digital signal processor that may also be used to provide additional processing power to augment the capabilities of the computing device **110**.

Processing unit **140** may include a central processing unit (CPU), a digital signal processing unit (DSP), a microprocessor, an application-specific integrated circuit (ASIC), a neural processing unit (NPU), a graphics processing unit (GPU), a field-programmable gate array (FPGA), and so forth. Processing unit **140** generally comprises a programmable processor that executes program instructions to manipulate input data. In some embodiments, processing unit **140** may include any number of processing cores, memories, and other modules for facilitating program execution.

For example, processor unit **140** could receive input from a user via input device(s) **174** and drive output device(s) **176** to emit soundwaves. In some embodiments, processing unit **140** can be configured to execute sound management application **130** in order to analyze acquired sensor data from sensor(s) **172** and modify the ambient sounds to be included in an output audio signal. In such instances, sound management application **130** may generate a modified ambient signal that removes selected ambient sound signals. Sound management application **130** may also generate, for each selected ambient sound signal selected for removal, a corresponding anti-noise signal that is the polar inverse of the selected ambient sound signal. In such instances, the anti-noise signal may cancel the selected ambient sound that bleeds through the output device (“bleeding sound”) such that the user does not hear the selected ambient sound, either as a portion of the audio signal reproduced by output device(s) **176**, or as a bleeding sound heard directly from the environment.

Memory **120** includes a memory module, or collection of memory modules. Memory **120** may include a variety of computer-readable media selected for their size, relative performance, or other capabilities: volatile and/or non-volatile media, removable and/or non-removable media, etc. Memory **120** may include cache, random access memory (RAM), storage, etc. Memory **120** may include one or more discrete memory modules, such as dynamic RAM (DRAM) dual inline memory modules (DIMMs). Of course, various memory chips, bandwidths, and form factors may alternately be selected.

Non-volatile memory included in memory **120** generally stores application programs including user interface **122** and/or sound management application **130**, and data (e.g., data stored in database **124**) for processing by processing unit **140**. In various embodiments, memory **120** may include non-volatile memory, such as optical drives, magnetic drives, flash drives, or other storage. In some embodiments, separate data stores, such as external data store **152** included in network **150** (“cloud storage”) may supplement memory **120**. Sound management application **130** within memory **120** can be executed by processing unit **140** to implement the overall functionality of computing device **110** and, thus, to coordinate the operation of transparent sound management system **100** as a whole.

In various embodiments, memory **120** may include one or more modules for performing various functions or techniques described herein. In some embodiments, one or more of the modules and/or applications included in memory **120** may be implemented locally on computing device **110**,

and/or may be implemented via a cloud-based architecture. For example, any of the modules and/or applications included in memory **120** could be executed on a remote device (e.g., smartphone, a server system, a cloud computing platform, etc.) that communicates with computing device **110** via network interface **142** or I/O devices interface **144**.

Sound management application **130** analyzes incoming sensor data and generates a composite ambient signal that attenuates and/or removes specific ambient sound signals that were included in the sensor data. In various embodiments, sound management application **130** may send a combined audio signal to output device(s) **176** that causes output device(s) **176** to emit soundwaves.

Sound management application **130** includes one or more modules to identify portions of acquired sensor data (e.g., acquired audio data that includes an input audio signal) and modify portions of the acquired sensor data to be included in an output audio signal. For example, as shown, sound management module **130** includes source identification module **132**, source separation module **134**, signal generation module **136**, and sound mixing module **138**.

Source identification module **132** identifies detected sound types, sound sources, and/or identifies “sound scenes” based on detected sounds. In some embodiments, source identification module **132** may execute various speech recognition, noise recognition, and/or other sound source identification techniques in order to identify portions of captured audio. For example, source identification module **132** could identify a specific sound scene, such as a park, by identifying different sound sources, including various bird calls, rustling leaves, running humans, and so forth. Additionally or alternatively, source identification module **132** may identify specific sound sources within a particular sound type. For example, source identification module **132** could identify two separate barking dogs within the input audio signal.

In some embodiments, source identification module **132** may store sound samples as sound data in database **124** and/or sound data **156** in external data store **152** and may identify a particular sound source by determining that a portion of the input audio signal matches stored sound data associated with a particular sound source. For example, a user could store the bark of a neighbor’s dog as sound data stored locally in database **124**. Source identification module **132** could then compare a portion of an input audio signal to the stored sound data. Upon determining that the portion of the input audio signal is substantially similar to the stored sound data, source identification module **132** could then identify the portion of the input audio signal as originating from the neighbor’s dog.

Source separation module **134** separates the input audio signal into separate component signals. In various embodiments, source separation module **134** may separate the input audio signal into separate signals based on the identified sounds and/or identified sound types that were determined by source identification module **132**. In such instances, source separation module **134** may generate distinct component audio signals from input audio signal such that each component audio signal may be separately modified. In some embodiments, source separation module **134** may separate the input audio signal into separate audio channels. In such instances, sound management application **130** may modify audio channel parameters (e.g., bandwidth, power level, etc.) in order to reproduce, amplify, attenuate, or remove a particular sound included in a specific audio channel.

Signal generation module **136** generates one or more audio signals to provide desired sound modifications. For example, signal generation module **136** could produce anti-noise signals that are scaled and/or polar inverse version of a given detected sound. In some embodiments, signal generation module **136** may also generate other waveforms in order to produce a desired sound modification. For example, signal generation module **136** may generate a periodic audio signals or random noise for use by one or more other modules.

Sound mixing module **138** combines multiple audio signals. For example, sound mixing module **138** could generate a modified ambient signal after selected ambient signals are removed by combining the remaining ambient signals. Similarly, sound mixing module **138** could generate a composite ambient signal by mixing the modified ambient signal with one or more anti-noise signals.

User interface **122** enables a user to provide input(s) about specific sound sources and/or sound types included in an ambient auditory environment that system **100** is to reproduce, amplify, attenuate, or remove. In various embodiments, user interface may provide other functions to enhance a sound; for example, user interface **122** may include equalization and/or filter controls. The user selections provided via user interface **122** are communicated to computing device **110** to control corresponding processing of input signals to create auditory output signals that implement the user's preferences. In some embodiments, user interface **122** may take any feasible form for providing the functions described herein, such as one or more buttons, toggles, sliders, dials, knobs, etc., or as a graphical user interface (GUI).

The GUI may be provided through any component of the transparent sound modification system **100**. In one embodiment, the GUI may be provided by a separate computing device that is communicatively coupled with computing device **110**, such as through an application running on a user's mobile or wearable computing device. To provide preferential selection of sound modification, user interface **122** may allow user input for various parameters such as direction(s), sound types, specific sound sources, and/or amount of sound modification to be performed. The parameters may be updated by the user or may be automatically updated during operation.

In another example, user interface **122** may receive verbal commands for user selections. In this case, computing device **110** may perform speech recognition on the received verbal commands and/or compare the verbal commands against commands stored in memory **120**. After verifying the received verbal commands, computing device **110** could then execute the commanded function for transparent sound modification system **100** (e.g., altering sound modification parameters to specified levels).

Database (DB) **124** may store values and other data retrieved by processing unit **140** to coordinate the operation of transparent sound modification system **100**. In various embodiments, in operation, processing unit **140** may be configured to store values in database **124** and/or retrieve values stored in database **124**. For example, database **124** may store sensor data, audio content, and reference audio (e.g., one or more reference audio signals) digital signal processing algorithms, transducer parameter data, and so forth.

In some embodiments, sound management application **130** may cause various signal characteristics of representative or average sounds of a particular sound group or sound type to be extracted and stored in database **124**. These signal

characteristics of representative sounds of a particular sound source or sound type may be used as a reference. In such instances, sound management application **130** (e.g., source identification module **132**) may compare sounds from the current ambient auditory environment with the reference sounds in order to identify the particular sound source or sound type. In some embodiments, specific sound references may be dynamically stored in database **124** based on the location of computing device **110**.

In some embodiments, database **124** may store other data acquired by computing device **110**, such as orientation data that reflects the relative orientation of sensor **172**, output device **176**, and/or computing device **110**. Additionally or alternatively, database **124** may store location data that reflects a geographic location of sensor **172**, output device **176**, and/or computing device **110**.

In some embodiments, computing device **110** may communicate with other devices, such as sensor(s) **172**, input device(s) **174**, and/or output device(s) **176**, using input/output (I/O) devices interface **144**. In such instances, I/O devices interface **144** may include any number of different I/O adapters or interfaces used to provide the functions described herein. For example, I/O devices interface **144** may include wired and/or wireless connections, and may use various formats or protocols. In another example, computing device **110**, through I/O devices interface **144**, could determine selected user inputs from input device(s) **174**, may detect ambient sounds using audio sensor(s) **172**, and may provide appropriate audio signals to output device(s) **176** to produce desired sound modifications for detected sounds. In another example, computing device **110** could, via I/O devices interface **144**, determine selected directions for sound modification using location data obtained from a separate computing device (e.g., a smartphone) that is connected using input device(s) **174**.

In some embodiments, computing device **110** may communicate with other devices, such as external data store **152** and/or audio source **160**, using network interface **142** through network **150**. In some embodiments, other types of networked computing devices (not shown) may connect to computing device **110** via network interface **142**. Examples of networked computing devices include a server, a desktop computer, a mobile computing device, such as a smartphone or tablet computer, and/or a worn device, such as a watch or headphones or a head-mounted display device. In some embodiments, the networked computing devices may be used as audio sensor(s) **172**, input device(s) **174**, and/or output device(s) **176**.

Network **150** includes a plurality of network communications systems, such as routers and switches, configured to facilitate data communication between computing device **110** and external data store **152**. Persons skilled in the art will recognize that many technically-feasible techniques exist for building network **150**, including technologies practiced in deploying an Internet communications network. For example, network **150** may include a wide-area network (WAN), a local-area network (LAN), and/or a wireless (Wi-Fi) network, among others.

External data store **152** includes sound libraries having stored sound data and/or other data use by computing device **110**. For example, external data store **152** could include environmental data **154** that computing sound management application **130** could use to identify the location of the ambient auditory environment and filter the set of reference sounds to which source identification module **132** references. In another example, external data store **152** could include sound data **156** that source identification module **132**

could access when identifying a particular sound source and/or sound type from the ambient audio environment.

Environmental data **154** includes data associated with the location of the ambient auditory environment. In various embodiments, environmental data **154** includes location data that specifies sound types and/or reference sounds associated with specific geographic locations and/or preset sound scenes (e.g., park, construction site, etc.). In various embodiments, a sound scene is an aggregation of sounds of one or more types associated with a certain setting. For example, a traffic sound scene could be an aggregation of sounds associated with street and/or highway traffic, such as sounds of cars travelling on a road, sounds of car horns, and so forth. In some embodiments, environmental data **154** may include direction data that reflects directions of sounds for sound modification within the ambient auditory environment.

Sound data **156** includes samples of individual sounds for use by source identification module **132** in identifying detected sounds. For example, sounds data **156** could include samples of sounds originating from cars, construction equipment, jackhammers, baby cries, human voices, barking dogs, and so forth. In some embodiments, sound data **156** may also include data used associated with reference sounds included in specific sound scenes.

Audio source **160** provides a source audio signal that computing device **110** sends to output device(s) **176** in order to be heard by the user. In various embodiments, audio source **160** may be a music player, an alert broadcaster, a stadium announcer, a store or theater, a conversation application (e.g., telephone application, video conferencing application, etc.) Streaming data may be provided directly from audio source **160** to the network interface **142** computing device **110** via network **150** through a wireless connection, such as Bluetooth® (a registered trademark of the Bluetooth Special Interest Group), Wi-Fi® (a registered trademark of the Wi-Fi Alliance) a cellular network, etc. Data streaming or downloads could also be provided over a local or wide-area network (WAN), such as the Internet, for example. In some embodiments, audio source may connect to computing device **110** via input device(s) **174** (e.g., a mobile device, a wired connection, etc.).

Sensor(s) **172** include one or more devices that collect data associated with objects in an environment. In various embodiments, sensor(s) **172** may include groups of sensors that acquire different sensor data. For example, sensor(s) **172** could include a reference sensor, such as a microphone and/or a position tracker (e.g., camera, thermal imager, linear position sensor, etc.), which could acquire sound data and/or location data (e.g., direction, relative position, etc.).

In various embodiments, sensor(s) **172** may include audio sensors, such as a microphone and/or a microphone array that acquires sound data. Such sound data may be processed by sound management application **130** using various audio processing techniques. The audio sensors may be a plurality of microphones or other transducers or sensors capable of converting sound waves into an electrical signal. The audio sensors may include an array of sensors that includes sensors of a single type, or a variety of different sensors. Sensor(s) **172** may be worn by a user, disposed separately at a fixed location, or movable. Sensor(s) **172** may be disposed in any feasible manner in the environment. In various embodiments, sensor(s) **172** are generally oriented outward relative to output device(s) **176**, which are generally disposed inward of sensor(s) **172** and also oriented inward. Such an orientation may be particularly beneficial for isolating one or more regions for which sound modification is to be performed (i.e., using output from the output device(s) **176**)

from the rest of the environment. In one example, sensor(s) **172** may be oriented radially outward from a user, while the output device(s) **176** are oriented radially inward toward the user.

Additionally or alternatively, sensor(s) **172** may include position sensors, such as an accelerometer or an inertial measurement unit (IMU). The IMU may be a device like a three-axis accelerometer, gyroscopic sensor, and/or magnetometer. In some embodiments, sensor(s) **172** may include optical sensors, such as RGB cameras, time-of-flight sensors, infrared (IR) cameras, depth cameras, and/or a quick response (QR) code tracking system. In addition, in some embodiments, sensor(s) **172** may include wireless sensors, including radio frequency (RF) sensors (e.g., sonar and radar), ultrasound-based sensors, capacitive sensors, laser-based sensors, and/or wireless communications protocols, including Bluetooth, Bluetooth low energy (BLE), wireless local area network (Wi-Fi) cellular protocols, and/or near-field communications (NFC).

Input device(s) **174** are devices capable of receiving one or more inputs. In various embodiments, input device(s) **174** may include one or more audio input devices, such as a microphone, a set of microphones (e.g., a reference microphone and an error microphone), and/or a microphone array. Additionally or alternatively, input device(s) **174** may include other devices capable of receiving input, such as a keyboard, a mouse, a touch-sensitive screen, and/or other input devices for providing input data to computing device **110**. For example, a user's input may include gestures, such as various movements or orientations of the hands, arms, eyes, or other parts of the body that are received via a camera.

Output device(s) **176** include devices capable of providing output, such as a display screen, loudspeakers, and the like. For example, output device **176** could be headphones, ear buds, a speaker system (e.g., one or more loudspeakers, amplifier, etc.), or any other device that generates an acoustic field. In various embodiments, various input device(s) **174** and/or output device(s) **176** can be incorporated into computing device **110**, or may be external to computing device **110**.

In various embodiments, output device(s) **176** may be implemented using any number of different conventional form factors, such as discrete loudspeaker devices, around-the-ear (circumaural), on-ear (supraaural), or in-ear headphones, hearing aids, wired or wireless headsets, body-worn (head, shoulder, arm, etc.) listening devices, body-worn close-range directional speakers or speaker arrays, body-worn ultrasonic speaker arrays, and so forth. In some embodiments, output device(s) **176** may be worn by a user, or disposed separately at a fixed location, or movable. As discussed above, output device(s) **176** may be disposed inward of the sensor(s) **172** and oriented inward toward a particular region or user.

Auditory Augmented Reality using Selective Noise Cancellation

FIG. 2 illustrates a technique of transparent sound modification system **100** of FIG. 1 applying active noise cancellation to a portion of an ambient audio signal, according to one or more embodiments. As shown active noise cancellation technique **200** includes sound management application **130**, audio sensor(s) **272**, and audio source **160**.

In operation, audio sensor(s) **272** acquires sensor data originating from sound sources **210** (e.g., **210(1)**, **210(2)**, **210(3)**) and transmit the acquired sensor data as input ambient signal **222** to sound management application **130**. Sound management application **130** processes and modifies

11

input ambient signal **222** to generate composite ambient signal **234**. Sound management application **130** combines composite ambient signal **234** with audio source signal **260**, originating from audio source **160**, to generate combined audio signal **270**. In operation, composite ambient signal **234** includes anti-noise signal **244** that, when reproduced as a soundwave, cancels bleeding ambient sound **212** originating from one of the sound sources (e.g., sound source **210(3)**) such that bleeding ambient sound **212** is not heard by the user.

Audio sensor(s) **272** acquire audio data from one or more sound sources **210** included in an environment, where the detected sounds form an ambient auditory environment. For example, the ambient auditory environment includes multiple sound sources **210**, including a barking dog **210(1)**, chirping bird **210(2)**, and an active jackhammer **210(3)**. Audio sensor(s) **272** sends the acquired audio data to sound management application **130** that processes the audio data as single input ambient signal **222**.

Sound identification module **132** analyzes input ambient signal **222** in order to identify individual sound sources **224**. For example, source identification module **132** could compare portions of input ambient signal **222** to one or more reference sounds stored in database **124** and/or external data store **152** in order to identify individual sound sources. In some embodiments, source identification module **132** may compare input ambient signal **222** to sounds within a reference sound scene in order to identify particular sound sources **210(1)-210(3)** that originated portions of input ambient signal **222**.

Sound separation module **134** separates input ambient signal **222** into multiple component ambient signals **228** (e.g., **228(1)**, **228(2)**, **228(3)**). As shown, for example, each component ambient signal corresponds to a distinct sound source, with component ambient signal **228(1)** originating from barking dog **210(1)**, component ambient signal **228(2)** corresponding to chirping bird **210(2)**, and component ambient signal **228(3)** corresponding to active jackhammer **210(3)**. In alternative embodiments, source separation module **134** may attempt to separate input ambient signal **222** into separate component signals, and source identification module **132** identifies the sound source **210** for each separate component ambient signal **228**.

In various embodiments, source separation module **134** may include each component ambient signal **228** in a separate audio channel **226**. For example, component ambient signal **228(3)** may be included in audio channel **226(3)**, which sound management application **130** processes independent of audio channels **226(1)**, **226(2)**. In some embodiments, sound management application **130** may dynamically generate user interface **122** based on the set of component ambient signals **228**. For example, sound management application **130** could cause user interface **122** to include separate controls corresponding to component ambient signals **228(1)**, **228(2)**, **228(3)**. In such instances, a user may provide separate controls of each component ambient signal. Alternatively, sound management application **130** may load a stored GUI that includes pre-defined sets of controls (e.g., an airport mode preset GUI).

Sound management application **130** may then modify each component ambient signal **228** based on selections made by the user via user interface **122**. For example, a user could opt to remove the active jackhammer **210(3)** from the ambient sounds that she hears. In such instances, sound management application **130** may use sound mixing module **138** to remove the signal from audio channel **226** (as illustrated by the empty box). In some embodiments, sound

12

management application **130** may apply active cancellation to component ambient signal **228(3)** by having signal generation module **136** generate a corresponding anti-noise signal that is used significantly attenuate or cancel the component ambient signal. Additionally or alternatively, sound management application **130** may modify parameters of audio channel **226(3)**, such as lowering the power of the audio channel. In some embodiments, sound mixing module **138** may amplify other component ambient signals **228**, such as amplifying component ambient signal **228(2)** (originating from chirping bird **210(2)**).

Sound management application **130** may then combine the remaining component ambient signals **228(1)**, **228(2)** to generate modified ambient signal **232**. In some embodiments, sound management application **130** may combine audio channels **226** into a single combined channel **230**. When mixing the remaining component ambient channels, the removed component ambient signal is not included.

In some embodiments, sound management application **130** may, in parallel, take audio sample **242** corresponding to each component ambient signal **228** that was removed. For example, sound management application **130** could generate audio sample **242** from component ambient signal **228(3)** that was removed from audio channel **226(3)**. In some embodiments, audio sample **242** is a reproduction of the removed component ambient signal **228(3)**. Alternatively, sound management application **130** may directly process the removed component ambient signal **228(3)** in lieu of generating a reproduction. Signal generation module **136** may generate an anti-noise signal **244** that is a polar inverse of audio sample **242**. In such instances, audio sample **242** and anti-noise signal **244** cancel when combined.

Upon generating anti-noise signal **244**, sound mixing module **138** combines anti-noise signal **244** with modified ambient signal **232** to generate composite ambient signal **234**. Composite ambient signal **234** includes each component ambient signal **228** that the user selected to hear (e.g., component ambient signals **228(1)**, **228(2)**), as well as anti-noise signals **244** corresponding to each component ambient signal **228(3)** that the user selected for removal.

Sound mixing module **138** generates combined audio signal **270** by mixing composite ambient signal **234** with an audio source signal **260** that is provided by audio source **160**. Because combined audio signal **270** includes both the audio source signal **260** and the composite ambient signal **234**, sound management application **130** drives an audio output device **176** to emit soundwaves that include both source audio and ambient sounds. Thus, the user can hear a selectively-transparent signal along with a desired audio source signal **260**, as at least a portion of the ambient auditory environment is reproduced by output device **176** emitting soundwaves based on combined audio signal **270**.

In various embodiments, one or more sound sources **210** may generate sounds that leak through various passive noise cancellation barriers. For example, when a user is wearing headphones, a sound source **210(3)** could generate sounds, represented by bleeding ambient sound **212**, that leak through passive noise cancellation barrier provided by the ear pad of the headphones and could potentially be heard by the user. In such instances, the anti-noise signal **244** included in composite ambient signal **234** and combined audio signal **270** cancels the bleeding ambient sound **222**. As a result, when output device **176** produces soundwaves based on combined audio signal **270**, at least a portion of the soundwaves provides active cancellation by canceling bleeding ambient sound **222** such that bleeding ambient sound **222** is not heard by the user.

FIG. 3 illustrates an example user interface **300** provided by the transparent sound modification system **100** of FIG. 1 to control portions of an ambient audio signal, according to one or more embodiments.

User interface **300** enables the user to create a preferential auditory experience by setting preferences with respect to what ambient sounds to hear, dim, amplify, or not hear at all. In operation, a user sets auditory preferences via user interface **300** that may be implemented by computing device **110** or by a second computing device, such as a smartphone, tablet computer, smartwatch, etc. In some embodiments, user preferences represented in user interface **300** may be associated with particular sound sources (e.g., dog (1) **330**, dog (2) **340**) and/or sound types, groups, or categories of sounds (e.g., construction **310**, voices **320**). Additionally or alternatively, user interface **300** and may include one or more modifications to the associated sound, such as cancellation (e.g., “off” setting **350**), attenuation (e.g., “low” setting **352**), reproduction (e.g., “real” setting **354**), and/or amplification (e.g., “loud” setting **356**) settings. Upon the user setting individual controls **542-548** corresponding to the distinct sound sources and/or sound types **310-340**, commands are sent to sound management application **130** in order to further process component ambient signals **228**.

In some embodiments, user interface **300** may be a context-sensitive or context-aware user interface. In such instances, sound management application **130** may first identify individual sound sources **224** and/or sound types and may also separate input ambient signal **222** into component ambient signals **228**. Sound management application **130** may then generate user interface **300** based on the identified sound sources **224** and/or the component ambient signals **228**.

When control **342** at the “off” setting **350**, sound management application **130** may then attenuate the associated component ambient signal such the user cannot hear the sound when listening to combined audio signal **270**. In some embodiments, sound management application **130** may apply active cancellation to the component ambient signal by generating a corresponding anti-noise signal that is used to significantly attenuate or cancel the component ambient signal. Additionally or alternatively, sound management application **130** may modify parameters of the audio channel corresponding to the component ambient signal, such as lowering the power of the audio channel.

The real setting **354** corresponds to substantially replicating the sound level from the ambient auditory environment to the user, such as if the user is not wearing a device that attenuates ambient sounds. In such instances, sound management application **130** may process the component ambient signal without significant attenuation or amplification.

The low setting **352** corresponds to some attenuation, or relatively lower amplification, of a given component ambient signal relative to the component ambient signals represented by user interface **300**. The loud setting **356** corresponds to amplification of a component ambient signal relative to other sounds and/or the level of that sound in the ambient auditory environment.

In other embodiments, user interface **300** controls that enable a user to specify sound levels or sound pressure levels (SPL) for individual audio channels. For example, user interface **300** could include controls to specify percentages of an initial loudness value a particular audio channel. In some embodiments, user interface **300** may specify sound level values as sound pressure levels (dBA SPL) and/or attenuation/gain values (e.g., specified in decibels). For example, a user may move a selector, to an attenuation value

of -20 decibels (dB) (e.g., corresponding to a low setting **352**) when the user would like to attenuate a particular sound. Further, the user may move a selector to a value of 0 dB (e.g., corresponding to the real setting **354**) when the user would like to pass-through a particular sound. In addition, the user may move a selector toward a gain value of $+20$ dB (e.g., corresponding to the loud setting **356**) when the user would like to enhance a particular sound by increasing the loudness of the sound.

In various embodiments, the individual settings **310-340** may be directed to more specific sound types or sound classes. For example, sound management application **130** may provide user interface **300** that includes individual controls, such as main voices, other voices, or TV voices. Similarly, controls for noises may include sub-controls or categories for birds, traffic, construction, machinery, airplane, etc.

FIG. 4 illustrates selection of directions within an environment for the transparent sound modification system **100** of FIG. 1 to modify, according to one or more embodiments. Although one particular embodiment including headphones is depicted, various alternative implementations (e.g., directional speakers towards user **425**) are also possible.

Environment **400** provides a top-down depiction of a user **425** wearing headphones **427** on her head. The user **425** has orientation **405** within the environment **400**. Though a simplified 2D representation of the user orientation and the environment is illustrated, the same principles would also apply to a 3D representation (e.g., capturing whether user **425** is leaning forward, back, to the left side, to the right side, etc.). Audio sensing zone **410** represents an overall sensing region within the ambient environment that extends from the microphones into a threshold distance in all directions. In some embodiments, audio sensing zone **410** represents the composite sensing regions of the various microphones included with headphones **427**. Depending on the operational mode of sound management application **130**, sounds detected by headphones **427** as coming from pass-through area **415** within the audio sensing zone **410** are permitted to pass through to user **425** without applying an active sound modification. In contrast, sounds detected by headphones **427** as coming from modification region **420** within audio sensing zone **410** are modified by sound management application **130** based on user settings.

In various embodiments, user **425** may select specific direction(s) and/or region(s) for sound modification. In one example, user **425** might select modification region **420** to be attenuated or amplified (e.g., corresponding to one of the ear cups of the headphones **427**). Alternatively, user **425** might specify an angle and angular width (say a center angle of 90° from the orientation **405**, with a 180° width), or multiple angles (from 0° - 180°). In such instances, sounds originating from sound source **412** may be attenuated by sound management application **130**, while sounds originating from sound source **414** are not altered by sound management application **130**.

In some embodiments, sound management application **130** may apply active noise cancellation for a component ambient signal corresponding to sound source **412** by generating an anti-noise signal **244** that is to be included in a combined audio signal **270** reproduced by headphones **427**. In such instances, the anti-noise signal **244** cancels the component ambient signal and user **425** does not hear sound source **412** even if a bleeding ambient sound **212** leaks through the passive noise cancellation effects provided by headphones **427**.

15

In some embodiments, user **425** may provide selections of specific directions and/or regions for modification through inputs, such as mechanical controls, voice controls, physical gestures, GUI controls, and so forth. For example, each ear pad of headphones **427** may include one or more buttons that user **425** presses to selectively apply sound modifications for one or more specific directions and/or regions. The type and amount of sound modification may vary for different modification areas.

In addition to directions, a user may also specify that certain frequency ranges are to be modified. For example, sound management application **130** may apply active noise cancellation for non-periodic sounds by attenuating specific frequency values associated with a sound source. In some embodiments, a user may select a pre-defined frequency range (corresponding to speech, automobile traffic, or other common noise source ranges) that sound management application **130** will attenuate by applying a bandpass filter that attenuates the frequency range. In such instances, sound management application **130** may apply the bandpass filter in lieu of generating an anti-noise signal that is based on the non-periodic signal.

In some embodiments, the modification area(s) specified by the user may track the user's orientation, or may remain fixed despite changes to the user's orientation. For example, the user could select for cancellation all sound originating from her right side. When sound management application is in an operating mode that tracks the user, the modification area shifts with the user such that sounds coming from the user's right side at any instant (even if the user has moved) will continue to be modified. In some embodiments, sound management application **130** may estimate the locations of specific noise sources that generated the detected sounds. For example, sound management application **130** may use a known coordinate system, such as Cartesian, polar, or spherical coordinates, to specify locations of sound sources **412**, **414** relative to user **425**.

Environment **430** also provides a top-down depiction of the user **425**. As shown, user has orientation **405**, and wishes to select a specific region for modification area **450** (e.g., an area located behind the user). User **425** then sets one or more modification areas **450** and one or more pass-through areas **415** within audio sensing zone **410**. For example, user **425** could select the directions by specifying particular angles. In an alternate embodiment, the user may specify a direction or particular angle, along with a modifier to describe the relative width of the modification area **450** (e.g., "narrow," "moderate," "wide").

Environment **460** also provides a top-down depiction of user **425**, where user **425** specifies directions for two different modification areas **480(1)**, **480(2)**. Setting the modification areas **480(1)**, **480(2)** may also operate to define one or more pass-through areas **470(1)**, **470(2)** within the audio sensing zone **410**. In various embodiments, user **425** may specify the angles, or ranges of angles, for each modification area **480(1)**, **480(2)**, which may be selected simultaneously or at different times.

FIG. 5 is a flow diagram of method steps to apply active noise cancellation to a portion of an ambient audio signal, according to one or more embodiments. Although the method steps are described with respect to the systems of FIGS. 1-4, persons skilled in the art will understand that any system configured to perform the method steps, in any order, falls within the scope of the various embodiments. In some embodiments, transparent sound modification system **100** may continually execute method **500** on captured audio in real-time.

16

As shown, method **500** begins at step **501**, where transparent sound modification system **100** acquires ambient sounds from an environment. In various embodiments audio sensor(s) **272** acquire audio data originating from various sound sources **210** within an environment and generates an input ambient signal **222** from the acquired sensor data.

At step **503**, transparent sound modification system **100** identifies sound sources in input ambient signal **222**. In various embodiments, source identification module **132** included in sound management application **130** analyzes input ambient signal **222** in order to identify individual sound sources **224**. For example, source identification module **132** could compare portions of input ambient signal **222** to one or more reference sounds stored in database **124** and/or external data store **152** in order to identify individual sound sources.

At step **505**, transparent sound modification system **100** separates input ambient signal **222**, based on sound sources, into component ambient signals **228**. In some embodiments, source separation module **134** separates input ambient signal **222** into multiple component ambient signals **228** (e.g., **228(1)**, **228(2)**, **228(3)**). For example, source separation module **134** could use the identified sound sources **210** to generate a set of component ambient signals **228**, where each component ambient signal **228** corresponds to a distinct sound source. In some embodiments, source separation module **134** may have each component ambient signal **228** included in a separate audio channel **226**. For example, component ambient signal **228(2)** could be included in audio channel **226(2)**, which sound management application **130** processes independent of audio channel **226(1)**.

At step **507**, transparent sound modification system **100** receives one or more user selections. In some embodiments, sound management application **130** may dynamically generate user interface **122** based on the set of component ambient signals **228**. In such instances, a user may provide selects to modify each component ambient signal **228**. Sound management application **130** may then receive the user selections and modify each component ambient signal **228** based on selections made by the user via user interface **122**.

At step **509**, transparent sound modification system **100** determines whether to remove a component ambient signal **228**. When sound management application **130** determines that the user selected the component ambient signal **228** for removal, transparent sound modification system **100** proceeds to step **511**; otherwise, sound management application **130** determines that the user did not select the component ambient signal **228** for removal and proceeds to step **511**.

At step **511**, transparent sound modification system **100** removes the selected component ambient signal **228** from input ambient signal **222**. For example, a user could opt to remove one sound **228(3)** from a set of ambient sounds that she hears. In such instances, sound management application **130** may use sound mixing module **138** to remove the component ambient signal **228**. In some embodiments, sound management application **130** may apply active cancellation to the component ambient signal **228** by having signal generation module **136** generate a corresponding anti-noise signal that is used significantly attenuate or cancel the component ambient signal. Additionally or alternatively, sound management application **130** may modify parameters of audio channel **226(3)**, such as lowering the power of the audio channel.

At step **513**, transparent sound modification system **100** generates an anti-noise signal. In various embodiments, sound management application **130** may generate an anti-

17

noise signal **244** that is a polar inverse of the selected component ambient signal **228**. In such instances, the anti-noise signal **244** would cancel an acoustic sound similar to the selected component ambient signal **228** when combined. For example, transparent sound modification system **100** could generate anti-noise signal **244** in anticipated of a bleeding acoustic sound that has characteristics similar to selected component ambient signal **228**. In such instances, the anti-noise signal **244** would combine with the bleeding ambient sound to cancel the bleeding acoustic sound. In various embodiments, transparent sound modification system **100** may add anti-noise signal **244** to remove selected component ambient signal **228** from input audio signal, such as in step **511**.

At step **515**, transparent sound modification system **100** does not remove a given component ambient signal. Upon determining at step **509** that the user did not select the given component ambient signal for removal, sound management application **130** may reproduce or amplify the given component ambient signal. At step **517**, transparent sound modification system **100** determines whether each component ambient signal has been processed. If not, transparent sound modification system **100** returns to step **509** to process another component ambient signal. Otherwise, transparent sound modification system **100** proceeds to step **519**.

At step **519**, transparent sound modification system **100** generates a composite ambient signal **234**. In various embodiments, sound mixing module **138** may mix the component ambient signals **228** that were not removed and the anti-noise signal **244** in order to generate composite ambient signal **234**. In some embodiments, sound mixing module **238** may first combine the remaining component ambient signals **228** to generate a modified ambient signal **232**. Sound mixing module **238** may then combine anti-noise signal **244** with modified ambient signal **232** to generate composite ambient signal **234**. Composite ambient signal **234** includes each component ambient signal **228** that the user selected to hear, as well as the anti-noise signals **244** corresponding to each component ambient signal that the user selected for removal.

At step **521**, transparent sound modification system **100** may optionally combine the composite ambient signal **234** with an audio source signal **260** to generate combined audio signal **270**. At step **523**, transparent sound modification system **100** drives output device **176** with composite ambient signal **234** or combined audio signal **270**. In various embodiments, due to combined audio signal **270** including both the audio source signal **260** and the composite ambient signal **234**, sound management application **130** drives an audio output device **176** to emit soundwaves that include both source audio and ambient sounds. Thus, the user can hear a selectively-transparent signal, as at least a portion of the ambient auditory environment is reproduced by output device **176** emitting soundwaves based on combined audio signal **270**.

In various embodiments, one or more sound sources **210** may generate sounds that leak through various passive noise cancellation barriers. For example, when a user is wearing headphones, a sound source **210(3)** could generate sounds, represented by bleeding ambient sound **212**, that leak through passive noise cancellation barrier provided by the ear pad of the headphones and could potentially be heard by the user. In such instances, the anti-noise signal **244** included in composite ambient signal **234** and combined audio signal **270** cancels the bleeding ambient sound **222**. As a result, when output device **176** produces soundwaves based on combined audio signal **270**, at least a portion of the sound-

18

waves provides active cancellation by canceling bleeding ambient sound **222** such that bleeding ambient sound **222** is not heard by the user.

In sum, a transparent sound modification system includes audio sensors arranged to detect sounds within an environment, and one or more output devices that provide a modified audio signal to a user. A sound management application included in a computing device of the transparent sound modification system operates to detect sounds within the environment and generate a modified audio signal that attenuates or removes at least one component signal corresponding to the detected sounds and to generate a modified ambient signal. The processing unit also generates an anti-noise signal based on the detected sound that is attenuated or removed in order to provide active noise cancellation. The processing unit includes the anti-noise signal in the modified audio signal used to drive an output device.

The processing unit drives the output device using the modified audio signal to cause the output device to emit soundwaves. Because the modified audio signal includes the anti-noise signal, a portion of the soundwave emitted by the output device cancels with any bleeding sound originating from the sound the user selected for removal. The user hears a soundwave that includes only detected sounds that the user did not select for removal.

One technological advantage of the disclosed approach relative to the prior art is that the user can select specific ambient sounds to hear when listening to a particular source signal. By removing specific component signals from an incoming ambient sound signal, the system may provide a selective acoustic transparency by reproducing only portions of an ambient acoustic environment that a user desires to hear. Further, by executing active noise cancellation on selected sounds, the disclosed approach enables a system to prevent a user from hearing particular sounds by cancelling bleeding sounds that are not removed using other noise cancellation techniques. These technical advantages provide one or more technological advancements over prior art approaches.

1. In various embodiments, a computer-implemented method comprises receiving an input signal representing an ambient auditory environment of a user, generating, from the input signal, a set of ambient audio signals that includes a first component signal and a second component signal, generating, based on the first component signal, a first inverse signal that is a polar inverse of the first component signal, removing the first component signal from the set of ambient audio signals to generate a modified set of ambient audio signals that includes at least the second component signal, generating a first composite signal that includes at least the first inverse signal and the modified set of ambient audio signals, and driving an audio output device to produce soundwaves based on the first composite signal.
2. The computer-implemented method of clause 1, further comprising receiving an audio source signal from an audio source, and combining the audio source signal with the first composite signal.
3. The computer-implemented method of clause 1 or 2, where removing the first component signal comprises adding a first copy of the first inverse signal to the modified set of ambient audio signals to generate a modified ambient signal, and generating the first composite signal comprises combining a second copy of the first inverse signal with the modified ambient signal.

19

4. The computer-implemented method of any of claims 1-3, where the first component signal is included in a first channel, the second component signal is included in a second channel, and removing the first component signal comprises attenuating the first channel below a defined threshold. 5
5. The computer-implemented method of any of claims 1-4, where the first component signal is associated with a first sound type in a plurality of source types, and the second component signal is associated with a second sound type in the plurality of source types. 10
6. The computer-implemented method of any of claims 1-5, further comprising comparing the first component signal to a plurality of audio recordings, where each audio recording included in the plurality of audio recordings is associated with a sound source type, identifying a first audio recording that is substantially similar to the first component signal, determining that a sound source type associated with the first audio recording is the first sound type, and associating the first component signal with the first sound type. 15 20
7. The computer-implemented method of any of claims 1-6, further comprising determining a first direction of a first sound source within the ambient auditory environment, and determining a second direction of a second sound source within the ambient auditory environment. 25
8. The computer-implemented method of any of claims 1-7, where removing the first component signal comprises determining a first frequency range that includes the first component signal, and applying a bandpass filter on the input signal. 30
9. The computer-implemented method of any of claims 1-8, where the first component signal is a non-periodic signal. 35
10. The computer-implemented method of any of claims 1-9, where the input signal includes a plurality of sound sources of a first sound type.
11. In various embodiments, a system comprises at least one audio sensor that acquires sound from an environment of a user and produces an input signal, an audio output device, and at least one processor coupled to the at least one audio sensor that performs the steps of receiving the input signal, generating, from the input signal, a set of ambient audio signals that includes a first component signal and a second component signal, generating, based on the first component signal, a first inverse signal that is a polar inverse of the first component signal, removing the first component signal from the set of ambient audio signals to generate a modified set of ambient audio signals that includes at least the second component signal, generating a first composite signal that includes at least the first inverse signal and the modified set of ambient audio signals, and driving the audio output device to produce sound-waves based on the first composite signal. 40 45 50 55
12. The system of clause 11, where the at least one audio sensor includes an array of audio sensors.
13. The system of clause 11 or 12, where the at least one audio sensor and the audio output device are included in a set of headphones. 60
14. The system of any of clauses 11-13, further comprising an output device, where the processor further performs the steps of in response to generating the set of ambient audio signals, generating a context-sensitive user interface that displays separate controls corresponding to the first component signal and the second

20

- component signal, and causing the output device to provide the context-sensitive user interface.
 15. The system of any of clauses 11-14, further comprising an output device, where the processor further performs the steps of determining a location of the environment, and loading a pre-defined user interface that is associated with the location, assigning separate controls corresponding to the first component signal and the second component signal, and causing the output device to provide the pre-defined user interface.
 16. The system of any of clauses 11-15, further comprising receiving an audio source signal from an audio source, and combining the audio source signal with the first composite signal.
 17. The system of any of clauses 11-16, where removing the first component signal comprises adding a first copy of the first inverse signal to the modified set of ambient audio signals to generate a modified ambient signal, and generating the first composite signal comprises combining a second copy of the first inverse signal with the modified ambient signal.
 18. In various embodiments, one or more non-transitory computer-readable media include instructions that, when executed by one or more processors, cause the one or more processors to perform the steps of receiving an input signal representing an ambient auditory environment of a user, generating, from the input signal, a set of ambient audio signals that includes a first component signal and a second component signal, generating, based on the first component signal, a first inverse signal that is a polar inverse of the first component signal, removing the first component signal from the set of ambient audio signals to generate a modified set of ambient audio signals that includes at least the second component signal, generating a first composite signal that includes at least the first inverse signal and the modified set of ambient audio signals, and driving an audio output device to produce sound-waves based on the first composite signal.
 19. The one or more non-transitory computer-readable media of clause 18, further including instructions that, when executed by the one or more processors, cause the one or more processors to perform the steps of receiving an audio source signal from an audio source, and combining the audio source signal with the first composite signal.
 20. The one or more non-transitory computer-readable media of clause 18 or 19, where removing the first component signal comprises adding a first copy of the first inverse signal to the modified set of ambient audio signals to generate a modified ambient signal, and generating the first composite signal comprises combining a second copy of the first inverse signal with the modified ambient signal.
- Any and all combinations of any of the claim elements recited in any of the claims and/or any elements described in this application, in any fashion, fall within the contemplated scope of the present invention and protection.
- The descriptions of the various embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments.
- Aspects of the present embodiments may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of

21

an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a "module" or "system." In addition, any hardware and/or software technique, process, function, component, engine, module, or system described in the present disclosure may be implemented as a circuit or set of circuits. Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

Aspects of the present disclosure are described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine. The instructions, when executed via the processor of the computer or other programmable data processing apparatus, enable the implementation of the functions/acts specified in the flowchart and/or block diagram block or blocks. Such processors may be, without limitation, general purpose processors, special-purpose processors, application-specific processors, or field-programmable gate arrays.

The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of

22

blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

While the preceding is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A computer-implemented method comprising:

receiving an input signal capturing an ambient auditory environment of a user;

separating the input signal into a set of audio signals that includes a first component signal of a first sound type and a second component signal of a second sound type; and

in response to generating the set of audio signals:

generating a context-sensitive user interface that concurrently displays a plurality of first controls and a plurality of second controls, the plurality of first controls being independent from the plurality of second controls;

receiving, from the user, a selection to remove the first component signal; and

in response to receiving the selection to remove the first component signal:

removing, using a first noise cancellation technique, the first component signal from the set of audio signals to generate a modified set of audio signals; determining that a portion of the removed first component signal bleeds through an audio output device as a bleeding acoustic sound;

removing the bleeding acoustic sound using a second noise cancellation technique different from the first noise cancellation technique, wherein the second noise cancellation technique comprises generating an anti-noise signal to cancel the bleeding acoustic sound; and

driving the audio output device to generate sound based on the modified set of audio signals and the anti-noise signal.

2. The computer-implemented method of claim 1, wherein: the plurality of first controls comprises controls for canceling the first component signal, attenuating the first component signal, reproducing the first component signal unchanged, and amplifying the first component signal.

3. The computer-implemented method of claim 1, wherein the plurality of second controls comprises a continuum of sound pressure levels or a continuum of gain levels for adjusting a loudness of the second component signal.

4. The computer-implemented method of claim 1, further comprising:

receiving, from the user, a selection to attenuate the second component signal; and

in response to receiving the selection to attenuate the second component signal, reducing a power level of the second component signal in the modified set of audio signals prior to driving the audio output device based on the modified set of audio signals.

5. The computer-implemented method of claim 1, further comprising:

receiving, from the user, a selection to amplify the second component signal; and

23

in response to receiving the selection to amplify the second component signal, increasing a power level of the second component signal in the modified set of audio signals prior to driving the audio output device based on the modified set of audio signals.

6. The computer-implemented method of claim 1, further comprising:

comparing the first component signal to a plurality of audio recordings, wherein each audio recording included in the plurality of audio recordings is associated with a sound source type;

identifying a first audio recording that is substantially similar to the first component signal;

determining that a sound source type associated with the first audio recording is the first sound type; and associating the first component signal with the first sound type.

7. The computer-implemented method of claim 1, wherein the first component signal comprises first ambient sounds originating from a first direction, and wherein the first direction is determined based on a direction and an angular width specified by the user.

8. The computer-implemented method of claim 1, wherein generating the modified set of audio signals further comprises:

combining a first audio channel associated with the first component signal and a second audio channel associated with the second component signal into a combined audio channel that does not include the removed first component signal.

9. The computer-implemented method of claim 1, wherein the first noise cancellation technique comprises applying active noise cancellation to non-periodic sounds included in the first component signal.

10. The computer-implemented method of claim 1, wherein the first noise cancellation technique comprises attenuating specific frequency values of the first sound type.

11. The computer-implemented method of claim 1, wherein the first noise cancellation technique comprises applying a bandpass filter to attenuate a pre-defined frequency range associated with the first sound type.

12. A system comprising:

at least one audio sensor that acquires sound from an environment of a user and produces an input signal; an audio output device; and

a first computing device comprising one or more processors coupled to the at least one audio sensor that performs the steps of:

receiving the input signal;

separating the input signal into a set of audio signals that includes a first component signal of a first sound type and a second component signal of a second sound type; and

in response to generating the set of audio signals:

generating a context-sensitive user interface that concurrently displays a plurality of first controls and a plurality of second controls, the plurality of first controls being independent from the plurality of second controls;

receiving, from the user, a selection to remove the first component signal; and

in response to receiving the selection to remove the first component signal:

removing, using a first noise cancellation technique, the first component signal from the set of audio signals to generate a modified set of audio signals;

24

determining that a portion of the removed first component signal bleeds through the audio output device as a bleeding acoustic sound;

removing the bleeding acoustic sound using a second noise cancellation technique different from the first noise cancellation technique, wherein the second noise cancellation technique comprises generating an anti-noise signal to cancel the bleeding acoustic sound; and

driving the audio output device to generate sound based on the modified set of audio signals and the anti-noise signal.

13. The system of claim 12, wherein the at least one audio sensor includes an array of audio sensors or a set of headphones.

14. The system of claim 12, wherein the one or more processors further perform the step of providing the context-sensitive user interface to a second computing device for presentation to the user, the second computing device being separate from the first computing device.

15. The system of claim 12, wherein generating the context-sensitive user interface comprises:

determining a location of the environment, and loading the context-sensitive user interface that is associated with the location.

16. The system of claim 12, wherein the one or more processors further perform the steps of:

receiving an audio source signal from an audio source; and

adding the audio source signal to the modified set of audio signals prior to driving the audio output device based on the modified set of audio signals.

17. The system of claim 12, wherein the context-sensitive user interface comprises a continuum of levels for adjusting a loudness of the second component signal, and wherein the one or more processors further perform the steps of:

receiving, from the user via the context-sensitive user interface, a first level from the continuum of levels; and amplifying or reducing a power level of the second component signal based on the first level.

18. One or more non-transitory computer-readable media including instructions that, when executed by one or more processors, cause the one or more processors to perform steps comprising:

receiving an input signal capturing an ambient auditory environment of a user;

separating the input signal into a set of audio signals that includes a first component signal of a first sound type and a second component signal of a second sound type; and

in response to generating the set of audio signals:

generating a context-sensitive user interface that concurrently displays a plurality of first controls and a plurality of second controls, the plurality of first controls being independent from the plurality of second controls;

receiving, from the user, a selection to remove the first component signal; and

in response to receiving the selection to remove the first component signal:

removing, using a first noise cancellation technique, the first component signal from the set of audio signals to generate a modified set of audio signals; determining that a portion of the removed first component signal bleeds through an audio output device as a bleeding acoustic sound;

25

removing the bleeding acoustic sound using a second
noise cancellation technique different from the
first noise cancellation technique, wherein the
second noise cancellation technique comprises
generating an anti-noise signal to cancel the bleed- 5
ing acoustic sound; and
driving the audio output device to generate sound
based on the modified set of audio signals and the
anti-noise signal.

19. The one or more non-transitory computer-readable 10
media of claim 18, wherein removing the first component
signal from the set of audio signals comprises applying a
bandpass filter to the set of audio signals.

20. The one or more non-transitory computer-readable 15
media of claim 18, wherein the steps further comprise:
generating, based on the first component signal, an
inverse signal that is a polar inverse of the first com-
ponent signal; and
adding the inverse signal to the modified set of audio
signals prior to driving the audio output device based 20
on the modified set of audio signals.

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26